



FedNeMo: Privacy-Preserving Federated Fine-Tuning of Nemotron for Healthcare

Model Finetuning & Customisation



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Instructions:

- *Teams can use more than one slide per section.*
- *Teams should restrict the total number of slides to a maximum of 15.*
- *Teams can add optional points which they believe would support their application.*
- *Please submit the presentations prior to the application deadline.*
- *Keep it simple and to the point for the reviewers to make judgement.*



TEAM INTRODUCTION

SKILLSET / EXPERIENCE



Team Introduction — Midnight Ciphers

Member	Focus area	Responsibility on FedNeMo
Aayush Deshpande	Federated systems / NVIDIA stack	FLARE integration, NeMo LoRA pipeline, system architecture
Laukik Rathod	Privacy & security ML	FedRand filter, Laplacian DP mechanism, RDP accounting
Harshal Yadav	Data engineering / FL pipelines	FedPS preprocessing, non-IID data partitioning, MTAE quality filter
Yash Kalal	Evaluation & visualisation	GIA/MIA evaluation protocol, Streamlit telemetry dashboard

- Domain focus: machine learning, federated learning, privacy-preserving systems, LLM fine-tuning.
- Building on NVIDIA NeMo 2.0 (LoRA/PEFT), NVIDIA FLARE (federated orchestration / DXO filters), and Nemotron.
- Motivated by the ABDM data-silo problem and the absence of a production-ready privacy layer for federated LLM fine-tuning.



AI Use Case

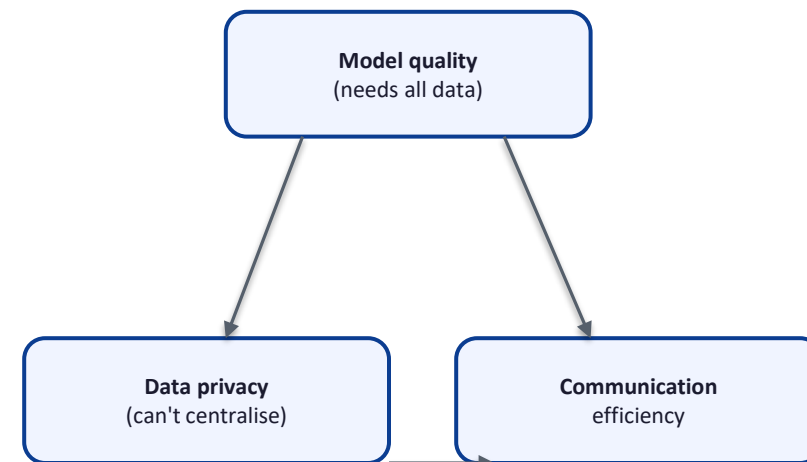
Expand on your project



AI Use Case — The Problem FedNeMo Solves

→ Project title: FedNeMo — Communication-Efficient, Privacy-Preserving Federated Fine-Tuning of Nemotron LLMs.

- India's ABDM has created 900 million health accounts and linked over 1 billion records. The data is digitised but siloed across hundreds of hospitals that cannot legally share it.
- DPDP Act 2023 designates hospitals as Data Fiduciaries; centralising patient data carries penalties up to ₹250 crore.
- Standard federated learning transmits LoRA weight updates — these can be inverted by a malicious server to reconstruct private patient records (Guo et al., Gradient Inversion Attacks).
- Adding noise destroys model quality. Cutting bandwidth hurts accuracy. No published system solves all three simultaneously for LLM fine-tuning.
- Motivation: we are actively building FedNeMo to fill this gap — a privacy-hardening layer that sits between FLARE and NeMo and makes federated Nemotron fine-tuning safe for regulated healthcare.



FedNeMo resolves all three.

AI Use Case — FedNeMo: How It Works

Component	Research basis	What it solves	Key result (from paper)
MTAE + OCSVM Data quality	Ardıç & Genç (IEEE Access 2025)	Removes poisoned / noisy samples before training	Up to 7.02% accuracy gain on non-IID data (CIFAR-10)
FedPS Preprocessing	Xu & Cormode (Feb 2026)	Harmonises lab values, ICD codes across hospitals without sharing data	Consistent preprocessing across heterogeneous FL clients
FedRand Privacy	Park et al. KAIST (2025)	Each round: only one LoRA matrix (A or B) transmitted — never both	Significant MIA resistance; accuracy maintained vs full LoRA
Adaptive Quant + Laplacian DP	Ardıç & Genç (IEEE Access 2025)	Formal $(\epsilon,0)$ -DP guarantee; entropy-driven compression	45–52% communication reduction; robust privacy on medical data
NVIDIA FLARE Orchestration	NVIDIA (FLARE 2.5+)	Federated round management; DXO filter pipeline for privacy	Production-grade FL transport; no refactoring of training loop

System flow: raw data never leaves the hospital — only processed LoRA deltas travel.



AI Use Case — Dataset Strategy & Evaluation Plan

Dataset	Use	Why this dataset	License
MedQA-USMLE	Primary training & evaluation	Medical QA; clean benchmark; simulates clinical reasoning	Research use (permissive)
Synthetic clinical records	Privacy-safe training data	Self-generated; no PHI risk; fully controllable non-IID distribution	No licence required
MIMIC-III / PubMedQA	Extended evaluation (planned)	Real EHR notes; rigorous clinical NLP benchmark	PhysioNet DUA / Apache 2.0

Evaluation plan — what we will measure:

- Non-IID setup: 5 simulated hospitals spanning ~100× data-volume range (metro tertiary → rural PHC) with differing ICD code distributions, driving entropy-weighted aggregation and per-client quantisation precision.
- Privacy metrics: run optimization-based Gradient Inversion Attack (DLG / InvertingGrad) and Membership Inference (LiRA) against (a) standard FedAvg and (b) FedNeMo. Report attack success rate before and after our defences.
- Utility metrics: MedQA accuracy across federated rounds; convergence vs centralised baseline; catastrophic-forgetting check on PIQA / ARC-Challenge.
- Efficiency: total bytes transmitted vs 32-bit FedAvg baseline across 50+ rounds.



Tech Stack

Strategy for building/improving your project



Tech Stack — NVIDIA Integration

Fine-tuning (PEFT)? Yes — LoRA on Nemotron via NVIDIA NeMo 2.0.

NVIDIA Technology	Why we need it	How FedNeMo uses it
NVIDIA FLARE 2.5+	Purpose-built federated orchestration; DXO filter pipeline is the exact extension point for client-side privacy filters	FedRandFilter, LaplacianDPFilter, AdaptiveQuantFilter as custom DXO Filter subclasses — no transport rewrite needed
NVIDIA NeMo 2.0	ModelTransform injects LoRA at Megatron-Core level, reaching Nemotron’s hybrid Mamba-2 + Transformer layers with mixed precision	llm.peft.LoRA targeting both attention and SSM projections; flare.patch(trainer) converts local trainer to federated client
Nemotron-Mini-4B (demo) / Nemotron-3 (prod)	Track-B required model; hybrid Mamba-Transformer handles 256K context — critical for longitudinal patient histories; multilingual (incl. Hindi) matches ABDM’s heterogeneous data; MoE routes specialties	Base model fine-tuned across 5 hospital clients; Adapter targets: linear_qkv, in_proj, out_proj, dt_proj
TensorRT-LLM / NIM	Post-training deployment of the federated model	Inference optimisation after training converges (planned)

Prior experience & tools: team working with NeMo framework documentation, FLARE Client API, HuggingFace transformers / PEFT. No prior production NVIDIA deployments — building toward that here.

Literature motivating approach: FedRand (Park et al.), Adaptive Quantization+DP (Ardıç & Genç), FedPS (Xu & Cormode), MTAE (Ardıç & Genç), GIA taxonomy (Guo et al.).

Tech Stack — Bottlenecks & How We Address Them

Potential bottleneck	Why it's a real risk	Our approach
DP noise destroys model quality	Naive Laplacian noise at strong ϵ collapses accuracy for clinical NLP tasks	FedRand transmits only one LoRA matrix per round — eliminates cross-matrix noise amplification ($O(1/\epsilon^4) \rightarrow O(1/\epsilon^2)$); entropy-weighted per-client ϵ keeps noise proportional to information value
Privacy budget grows unbounded across 100+ rounds	Naïve composition: 100 rounds $\times \epsilon=1.0$ gives $\epsilon_{total}=100$ — no meaningful privacy	Rényi DP (RDP) accounting with enforced budget ceiling; admin-configurable ϵ_{max} with hard training termination; live ϵ_{total} display on telemetry dashboard
Mamba-2 fused CUDA kernels may silently bypass LoRA adapters	NVIDIA's selective_scan_cuda and causal_conv1d_cuda call underlying weights directly, skipping Python-level hooks	Primary: NeMo 2.0 ModelTransform injects before kernel compilation; Fallback: pre/post-hook weight materialisation; Validation gate: asserts non-zero LoRA gradient norm before every run
Non-IID data drift across hospitals	Tertiary hospital (50K notes, 20 ICD codes) vs rural PHC (3K notes, 5 ICD codes) — naïve FedAvg diverges	Shannon-entropy client importance weighting in aggregation; FedPS harmonises structured metadata (labs, ICD, vitals) across inconsistent hospital EHR systems
FedPS scope: free-text clinical notes	FedPS was designed for tabular / structured data; discharge summaries are unstructured free text	FedPS handles structured EHR metadata; shared Nemotron tokenizer + canonical prompt template enforces consistency for free-text narratives



Road Map

Ideas for further improvement of the solution.



Roadmap — Build Plan & Mentor Support

Phase	Goal	Key deliverable
Phase 1 (Weeks 1–2)	End-to-end federated LoRA on Nemotron-Mini-4B	FLARE multi-site simulation running; NeMo LoRA fine-tuning on MedQA; baseline FedAvg accuracy established
Phase 2 (Weeks 3–4)	Privacy hardening	FedRand + Laplacian DP + RDP accounting implemented as FLARE DXO filters; live ϵ_{total} dashboard; GIA attack demonstration
Phase 3 (Weeks 5–6)	Communication efficiency + data quality	Adaptive quantisation scheduler; FedPS preprocessing; MTAE+OCSVM outlier filtering; non-IID partition evaluation
Phase 4 (Week 7)	Evaluation + submission	Privacy-utility curve; MIA study; convergence vs centralised baseline; Streamlit demo dashboard; final presentation

Limitations we are aware of:

- FedRand MIA numbers from Park et al. are on vision-language models — we will measure text-specific MIA on Nemotron.
- MTAE accuracy gains (7%) are from CIFAR-10 image classification — we will validate on clinical text.
- Nemotron-3 Nano (30B) is the production target; demo uses Mini-4B for hardware feasibility.

Mentor support requested:

- ① NeMo ModelTransform LoRA injection ordering vs Mamba-2 fused kernels.
- ② NVIDIA FLARE Secure Aggregation workflow integration.



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